PROJECT-BASED LEARNING

PHASE-II EVALUATION

Bus Route Optimizer

Team ID: T037

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T037
Semester: 3rd
Section: CSE
Domain: Data Structures & OOP

Team Members

1. Krishna Kant – 2510380211
2. Agrima Mishra – 2512510019
3. Aditi Singh – 2510012426
4. Akhil Gahlawat – 2510012465

Mentor

Dr.Vidit Kumar

Interactions completed:7

Phase-I Recap & Feedback Incorporated

Phase-I Status

basic idea and prototype was proposed in phase 1 .

Feedback Received

- Keep the project focused on core DSA and OOP concepts
- Clearly separate route finding from booking operations
- Validate the working features and document the results

Changes Incorporated

- Organized the system into route, management and booking modules
- Added BFS and Dijkstra route-search approaches
- Included file-based storage and Admin/Passenger menus

Problem Statement & Finalized Objectives

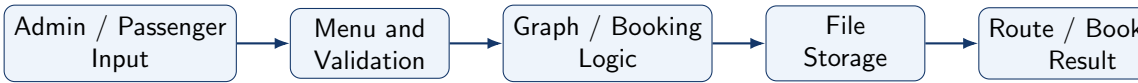
Problem Statement

To construct and simulate the c++ program of bus route optimizer.

Finalized Objectives

1. Represent bus stops and connections as a weighted graph
2. Use BFS to find routes with fewer stops and Dijkstra for shortest distance
3. Provide bus, stop, route and passenger management
4. Support booking, cancellation, seat handling and waiting lists

Updated System Architecture / Workflow



Major Components

- Graph and route-search module
- Bus, stop and route management
- Passenger booking and waiting-list management

Changes Since Phase-I

- Integrated route search with bus-management functions
- Added file-based persistence
- Connected Admin and Passenger console menus

Module Design & Functional Requirements

Major Modules

1. Route Finder
2. Bus Management
3. Bus Booking
4. Persistence and Accounts

Functional Requirements

- Add and manage buses, stops and routes
- Search for routes using BFS and Dijkstra
- Book or cancel tickets and manage seats
- Load and save records using files

Technology Stack & Development Methodology

Technology Stack

- Programming: C++
- Storage: Text files using fstream
- Framework: Console-based application
- Development Tool: C++ compiler / IDE
- Version Control: Git / GitHub (details to verify)

Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C	Graphs, adjacency lists, queues, vectors and priority queues	Route Finder
OOPs with C++	Classes, encapsulation, inheritance, virtual functions and STL containers	Bus Booking

Development & Implementation Progress

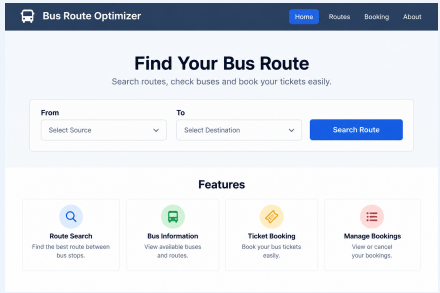
Module	Status	Work Completed
Route Finder	Completed	Graph representation, BFS and Dijkstra route search
Bus Management	Completed	Graph representation, BFS and Dijkstra route search

Working Prototype / Demonstration

Implemented Features

- Admin and Passenger console menus
- Bus, stop and route management
- BFS and Dijkstra route search
- Ticket booking, cancellation and waiting list

Prototype / Output



Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Route search	Valid route returned	Route displayed successfully for the selected source and destination	Pass
Ticket booking	Ticket booked if seats are available	Ticket booking completed successfully when seats were available	Pass
File persistence	Data is saved and loaded correctly	Data saving and loading tested	Pass

GitHub Repository & Development Activity

Repository Information

- GitHub Link: https://github.com/kkantsri/PBL-SEM-3

Individual Contribution

Member	Role	Technical Contribution
Krishna Kant	Team Lead	Graph implementation using adjacency lists, BFS and Dijkstra routing; system integration and over- all co-ordination.
Agrima Mishra	Team Member	Bus stop and route management; implementation of bus search, route creation, and network display features.
Aditi Singh	Team Member	Ticket booking and cancellation; seat availability, fare calculation, and waiting-list queue manage- men
Akhil Gahlawat	Team Member	File handling and data persistence; testing sys- tem features, checking edge cases, and preparing project documentation

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- Representing stops and route distances consistently
- Connecting route search, booking and file handling
- Keeping stored records and seat counts consistent

Solutions / Learning

- Separate the system into focused modules and test each function
- Applied graphs, queues, priority queues and OOP in C++

Phase-III Roadmap

1. Validate all implemented modules and edge cases
2. Fix integration issues and improve input validation
3. Run and record systematic tests
4. Demonstrate the working console application
5. Complete final report and presentation

Expected Outcomes & References

Expected Final Outcomes

- Working C++ console-based Bus Route Optimizer
- Test results for route search, booking and persistence
- Demonstrates core ideas used in route-planning systems
- Could be extended with a GUI, GPS or database later
- Academic demonstration of DSA and OOP integration

References

Cormen et al., *Introduction to Algorithms* Course notes on graph traversal and shortest paths Bjarne Stroustrup, *The C++ Programming Language* C++ reference documentation: `https://en.cppreference.com/` Project source code and test records

Thank You
Questions & Suggestions